

Optimization Methods

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Contents

1	Optimization Problems	1
1.1	Minimizers	1
1.2	Optimality Conditions	2
2	Line Search	3
2.1	Wolfe Conditions	3
2.2	Convergence of Line Search	4
3	Gradient Descent for Convex Functions	6
3.1	Convex Functions and Sublinear Convergence	6
3.2	Strong Convexity and Linear Convergence	8

1 Optimization Problems

1.1 Minimizers

Definition 1.1. (*Optimization Problem*)

Let f_0 be a real-valued objective and let x be the optimization variable. A problem with inequality and equality constraints is

$$\min_x f_0(x) \quad \text{s.t.} \quad \begin{cases} f_i(x) \leq 0 \\ h_j(x) = 0. \end{cases}$$

The functions f_i and h_j are the constraints, and $D = \{x \mid f_i(x) \leq 0, h_j(x) = 0\}$ is the feasible set.

Definition 1.2. (*Global and Local Minimizers*)

A point $x^* \in D$ is a global minimizer of $f : D \rightarrow \mathbb{R}$ if $\forall x \in D, f(x^*) \leq f(x)$. It is a local minimizer if some neighborhood N of x^* satisfies $\forall x \in N \cap D, f(x^*) \leq f(x)$. The minimizer is strict if the inequality is strict for every $x \neq x^*$.

Definition 1.3. (*Semicontinuity*)

A function f is lower semicontinuous at x^* if

$$f(x^*) \leq \liminf_{x \rightarrow x^*} f(x),$$

and upper semicontinuous at x^* if $f(x^*) \geq \limsup_{x \rightarrow x^*} f(x)$.

Definition 1.4. (*Minimizing Sequence*)

Let $f^* = \inf f(x)$. A sequence $\{x_k\}$ is minimizing if $f(x_k) \rightarrow f^*$.

Proposition 1.1.

Suppose f is lower semicontinuous on D . If a minimizing sequence $\{x_k\}$ has a cluster point $x^* \in D$, then $f(x^*) = f^*$. In particular, x^* is a global minimizer.

Proof.

Along a subsequence converging to x^* one still has $f(x_{k_j}) \rightarrow f^*$. Lower semicontinuity gives $f(x^*) \leq \liminf_j f(x_{k_j}) = f^*$. The reverse inequality holds because f^* is the infimum and $x^* \in D$. ■

Remark.

A minimizing sequence need not converge to a minimizer.

1.2 Optimality Conditions

Let $D \subset \mathbb{R}^d$ be open and let $f : D \rightarrow \mathbb{R}$.

Definition 1.5. (*Gradient*)

The function f is differentiable at $x \in D$ if there is a linear form ℓ such that

$$f(x + h) = f(x) + \ell(h) + o(\|h\|) \quad (h \rightarrow 0).$$

Then $\ell(h) = \langle \nabla f(x), h \rangle$, where

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_d} \right)^T$$

is the gradient of f at x .

Definition 1.6. (*Hessian*)

If the second partial derivatives exist, the Hessian $H(x) = \nabla^2 f(x)$ is the matrix

$$(\nabla^2 f(x))_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}.$$

If $f \in C^2$, then $\partial_{ij} f = \partial_{ji} f$, namely $\nabla^2 f$ is symmetric.

Theorem 1.1. (*Taylor Expansion*)

Suppose $f \in C^2$. Then there exists $t \in (0, 1)$ respectively such that

$$f(x + p) = f(x) + p^T \nabla f(x + tp),$$

$$f(x + p) = f(x) + \nabla f(x)^T p + \frac{1}{2} p^T \nabla^2 f(x + tp) p,$$

$$\nabla f(x + p) = \nabla f(x) + \int_0^1 \nabla^2 f(x + tp) p \, dt.$$

Theorem 1.2. (*Necessary Conditions*)

Suppose x^* is a local minimizer of f and $\nabla^2 f$ exists and is continuous on an open neighborhood of x^* . Then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semidefinite.

Theorem 1.3. (*Sufficient Conditions*)

Suppose $f \in C^2$ on a neighborhood of x^* . If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite, then x^* is a strict local minimizer of f .

2 Line Search

2.1 Wolfe Conditions

Definition 2.1. (*Descent Direction*)

A vector p is a descent direction for f at x if $p^T \nabla f(x) < 0$. For every sufficiently small $\varepsilon > 0$ one then has $f(x + \varepsilon p) < f(x)$.

Proposition 2.1.

Let $f(x) = \frac{1}{2} x^T A x - b^T x$ with A symmetric and positive semidefinite, and let $p^T A p > 0$. The exact line-search step from x along p is

$$\alpha = -\frac{\nabla f(x)^T p}{p^T A p}, \quad \nabla f(x) = Ax - b.$$

At a point x_k , fix a descent direction p_k and write $f_k = f(x_k)$, $\nabla f_k = \nabla f(x_k)$, and $\varphi(\alpha) = f(x_k + \alpha p_k)$ for $\alpha > 0$.

Definition 2.2. (*Wolfe Conditions*)

Fix constants $0 < c_1 < c_2 < 1$. A step $\alpha > 0$ satisfies the *Armijo* condition, or sufficient-decrease condition, if

$$f(x_k + \alpha p_k) \leq f_k + c_1 \alpha \nabla f_k^T p_k.$$

It satisfies the curvature condition if

$$\nabla f(x_k + \alpha p_k)^T p_k \geq c_2 \nabla f_k^T p_k,$$

namely $\varphi'(\alpha) \geq c_2 \varphi'(0)$. The two inequalities together are the *Wolfe* conditions.

Remark.

Since $\varphi'(0) < 0$, the Armijo inequality holds for every sufficiently small $\alpha > 0$. If $\varphi'(\alpha)$ is still largely negative, a longer step can decrease f further. The curvature condition excludes those short steps.

Proposition 2.2. (*Existence*)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable and p is a descent direction at x . If f is bounded from below on the ray $\{x + \alpha p \mid \alpha > 0\}$ and $0 < c_1 < c_2 < 1$, then the steps α that satisfy the *Wolfe* conditions form a nonempty union of open intervals.

Definition 2.3. (*Strong Wolfe Conditions*)

The strong *Wolfe* conditions keep the Armijo inequality and replace the curvature inequality by a two-sided bound: for some $c_2 \in (c_1, 1)$,

$$|\nabla f(x_k + \alpha p_k)^T p_k| \leq -c_2 \nabla f_k^T p_k.$$

Definition 2.4. (*Goldstein Condition*)

For a constant $c \in (0, 1/2)$, a step $\alpha > 0$ satisfies the *Goldstein* condition if

$$f_k + (1 - c)\alpha \nabla f_k^T p_k \leq f(x_k + \alpha p_k) \leq f_k + c\alpha \nabla f_k^T p_k.$$

2.2 Convergence of Line Search

Let θ_k be the angle between p_k and $-\nabla f_k$, so

$$\cos \theta_k = -\frac{\nabla f_k^T p_k}{\|\nabla f_k\| \|p_k\|}.$$

Descent directions have $\cos \theta_k > 0$.

Theorem 2.1. (*Zoutendijk*)

Suppose every p_k is a descent direction and every step α_k satisfies the Wolfe conditions. Assume f is bounded from below and continuously differentiable on an open set N containing $\{x : f(x) \leq f(x_0)\}$, and assume ∇f is Lipschitz continuous on N with constant L . Then

$$\sum_{k=0}^{\infty} \cos^2 \theta_k \|\nabla f_k\|^2 < \infty.$$

Proof.

The Armijo condition gives $f_{k+1} - f_k \leq c_1 \alpha_k \nabla f_k^T p_k$. The curvature condition rearranges to

$$(\nabla f_{k+1} - \nabla f_k)^T p_k \geq (c_2 - 1) \nabla f_k^T p_k.$$

Lipschitz continuity of the gradient bounds the same inner product by $L \alpha_k \|p_k\|^2$, so

$$\alpha_k \geq \frac{(c_2 - 1) \nabla f_k^T p_k}{L \|p_k\|^2} > 0.$$

Insert this lower bound into the decrease of f . Since $\nabla f_k^T p_k < 0$,

$$f_{k+1} - f_k \leq -\frac{c_1(1 - c_2)}{L} \cos^2 \theta_k \|\nabla f_k\|^2.$$

Summing the inequality yields

$$\frac{c_1(1 - c_2)}{L} \sum_{k=0}^{\infty} \cos^2 \theta_k \|\nabla f_k\|^2 \leq f(x_0) - \inf f < \infty.$$

■

Corollary 2.1.

Under the hypotheses of Zoutendijk's theorem, $\cos^2 \theta_k \|\nabla f_k\|^2 \rightarrow 0$. If $\cos \theta_k \geq \delta > 0$ for every k , then $\|\nabla f_k\| \rightarrow 0$.

3 Gradient Descent for Convex Functions

3.1 Convex Functions and Sublinear Convergence

Definition 3.1. (*Convex Set and Convex Function*)

A set D is convex if $\forall x, y \in D, \forall t \in [0, 1], tx + (1 - t)y \in D$. A function $f : D \rightarrow \mathbb{R}$ is convex if D is convex and

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \quad \alpha \in [0, 1].$$

Remark.

On the interior of its domain, a convex function is differentiable almost everywhere and Lipschitz on compact subsets. If f is differentiable, then ∇f is monotone: $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$, and Jensen's inequality holds.

Proposition 3.1.

Let $f : D \rightarrow \mathbb{R}$ be differentiable and convex. A point $x_0 \in D$ is a global minimizer if and only if $\nabla f(x_0) = 0$.

Proposition 3.2. (*Smoothness Inequalities*)

Suppose f is convex and differentiable and ∇f is Lipschitz continuous with constant M , namely $\nabla^2 f \preceq MI$ when the Hessian exists. Then

$$\frac{1}{2M} \|\nabla f(x) - \nabla f(y)\|^2 \leq f(y) - f(x) - \nabla f(x)^T(y - x) \leq \frac{M}{2} \|x - y\|^2,$$

and ∇f is monotone and cocoercive:

$$\frac{1}{M} \|\nabla f(x) - \nabla f(y)\|^2 \leq \langle \nabla f(x) - \nabla f(y), x - y \rangle \leq M \|x - y\|^2.$$

Remark.

If a minimizer x^* exists, the first inequality at $x = x^*$ reduces to

$$\frac{1}{2M} \|\nabla f(y)\|^2 \leq f(y) - f(x^*) \leq \frac{M}{2} \|y - x^*\|^2.$$

If $\nabla^2 f \preceq MI$, Taylor expansion along the gradient step gives

$$f(x_k - \alpha \nabla f_k) \leq f_k - \alpha \|\nabla f_k\|^2 + \frac{1}{2} \alpha^2 M \|\nabla f_k\|^2.$$

The right-hand side is minimized at $\alpha = 1/M$, and that choice leaves

$$f_{k+1} \leq f_k - \frac{1}{2M} \|\nabla f(x_k)\|^2.$$

The objective therefore decreases at every nonstationary point.

Theorem 3.1.

Apply gradient descent with constant step $\alpha_k = 1/M$ to a continuously differentiable convex function satisfying $\nabla^2 f \preceq MI$. If f is bounded from below, then $\nabla f(x_k) \rightarrow 0$. If at least one minimizer exists, then x_k converges to some minimizer.

Theorem 3.2. (Sublinear Rate)

Apply gradient descent with constant step $\alpha_k = 1/M$ to a continuously differentiable convex function satisfying $\nabla^2 f \preceq MI$. If a minimizer x^* exists, then for every integer $k \geq 1$,

$$f(x_k) - f(x^*) \leq \frac{2M \|x_0 - x^*\|^2}{k}.$$

Proof.

Write $\Delta_k = f(x_k) - f(x^*)$. The constant-step decrease and the tangent inequality $\nabla f_k^T(x_k - x^*) \geq \Delta_k$ give

$$\|\nabla f_k\| \geq \frac{\Delta_k}{\|x_k - x^*\|},$$

and therefore

$$\Delta_{k+1} \leq \Delta_k - \frac{\Delta_k^2}{2M \|x_k - x^*\|^2}.$$

The distance to x^* is at most $R = \|x_0 - x^*\|$, so

$$\Delta_j - \Delta_{j+1} \geq \frac{\Delta_j^2}{2MR^2}$$

for every j . If $\Delta_k = 0$, the rate is immediate. Otherwise $\Delta_j > 0$ for all $j \leq k$, and

$$\frac{1}{\Delta_{j+1}} - \frac{1}{\Delta_j} \geq \frac{1}{2MR^2}.$$

Summing from $j = 0$ to $k - 1$ produces

$$\frac{1}{\Delta_k} \geq \frac{1}{\Delta_0} + \frac{k}{2M \|x_0 - x^*\|^2},$$

and dropping the positive initial term yields the claimed rate. ■

3.2 Strong Convexity and Linear Convergence

Definition 3.2. (*Strong Convexity*)

A differentiable function f is strongly convex with modulus $m > 0$ if $x \mapsto f(x) - \frac{m}{2} \|x\|^2$ is convex. Equivalently, when the Hessian exists, $\nabla^2 f \succeq mI$.

Proposition 3.3.

Suppose f is differentiable, strongly convex with modulus m , and ∇f is Lipschitz continuous with constant M . At the minimizer x^* ,

$$\frac{1}{2M} \|\nabla f(x)\|^2 \leq f(x) - f(x^*) \leq \frac{1}{2m} \|\nabla f(x)\|^2$$

and

$$\frac{1}{M} \|\nabla f(x)\| \leq \|x - x^*\| \leq \frac{1}{m} \|\nabla f(x)\|.$$

Theorem 3.3. (*Improved Cocoercivity*)

Suppose f is differentiable, strongly convex with modulus m , and ∇f is Lipschitz continuous with constant M . Then

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{Mm}{M+m} \|x - y\|^2 + \frac{1}{M+m} \|\nabla f(x) - \nabla f(y)\|^2.$$

Theorem 3.4. (*Linear Rate*)

Suppose f is differentiable, strongly convex with modulus m , and ∇f is Lipschitz continuous with constant M . If the constant gradient step satisfies $0 < \alpha \leq 2/(M+m)$, then

$$\|x_k - x^*\|^2 \leq c^k \|x_0 - x^*\|^2, \quad c = 1 - \frac{2Mm\alpha}{M+m} \in [0, 1).$$

Consequently

$$f(x_k) - f(x^*) \leq \frac{M}{2} c^k \|x_0 - x^*\|^2.$$

Proof.

The iteration $x_{k+1} = x_k - \alpha \nabla f(x_k)$ expands as

$$\|x_{k+1} - x^*\|^2 = \|x_k - x^*\|^2 - 2\alpha \nabla f(x_k)^T (x_k - x^*) + \alpha^2 \|\nabla f(x_k)\|^2.$$

Improved cocoercivity at the pair (x_k, x^*) , where $\nabla f(x^*) = 0$, gives

$$\nabla f(x_k)^T (x_k - x^*) \geq \frac{Mm}{M+m} \|x_k - x^*\|^2 + \frac{1}{M+m} \|\nabla f(x_k)\|^2.$$

Therefore

$$\|x_{k+1} - x^*\|^2 \leq \left(1 - \frac{2Mm\alpha}{M+m}\right) \|x_k - x^*\|^2 - \left(\frac{2\alpha}{M+m} - \alpha^2\right) \|\nabla f(x_k)\|^2.$$

The second coefficient is nonnegative precisely when $0 \leq \alpha \leq 2/(M+m)$, so it may be dropped. The resulting contraction iterates to the bound on $\|x_k - x^*\|^2$. The estimate $f(x) - f(x^*) \leq (M/2) \|x - x^*\|^2$ from smoothness upgrades it to a bound on the function values. ■

Remark.

The factor c is decreasing in α on the admissible interval, so the smallest contraction available from this argument occurs at

$$\alpha = \frac{2}{M+m}, \quad c = \left(\frac{M-m}{M+m}\right)^2 = \left(\frac{\kappa-1}{\kappa+1}\right)^2, \quad \kappa = \frac{M}{m}.$$